

Integrated Planning of Distribution Networks Considering Utility Planning Concepts

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Abstract—This paper presents a methodology for integrated planning of real-life medium-voltage networks based on the utility planning concepts. The research is motivated by the need to develop a methodology that would line-up with utility day-to-day businesses and could be applied in real-life. Its core is a two-stage optimization process, where the first stage solves the static investment optimization and the second stage considers operational problem. A probabilistic decision tree approach is proposed for the solution of the entire problem to consider uncertainties in the planning period. The overall formulation is given first, which is followed by details of the investment model and outlines of the proposed operation planning. The novelty of the investment problem, which determines optimal network reinforcements, is explicit modeling of network security constraints of radially operated networks, whilst considering different operating regimes. Additional novel features include modeling of real-life supply restoration rules through network reconfiguration and optimal placement of new switching devices, as well as consideration of “customer flows” on the network. Connection of new distributed generation and demand centers and construction of circuits on new corridors are also included. Two investment models are formulated as mixed-integer nonlinear optimization problems, tested on several MV networks and compared with established methods. The proposed operational problem is solved in two stages, quality-of-supply and operation cost optimization. Computational aspects are also presented.

Index Terms—Distribution investment planning, medium-voltage networks, mixed-integer nonlinear optimization, optimal reconfiguration, security standard.

NOMENCLATURE

Sets

Ω_l	Set of all branches ($\Omega_l = \Omega_{lex} \cup \Omega_{lnew}$)
Ω_{lex}	Set of existing branches
Ω_{lnew}	Set of new branches (DG connections and new corridors)
Ω_{Con}	Set of new branches that can be used to connect DG
Ω_b	Set of all buses
Ω_{SW}	Set of all switchgear ($\Omega_{SW} = \Omega_{SWex} \cup \Omega_{SWnew}$)

Ω_{SWex}	Set of existing switchgear
Ω_{SWnew}	Set of new switchgear
$\Omega_{CB-RECnew}$	Set of new CBs/reclosers (operation stage)
$\Omega_{SWITnew}$	Set of new switches (operation stage)
Ω_{NOP}	Set of all normally open switches to adjacent feeders
Ω_k	Set of intact/non – intact regimes ($\Omega_k = \Omega_k^{nor} \cup \Omega_k^{abn}$)
Ω_k^{nor}	Set of intact regimes
Ω_k^{abn}	Set of non—intact (contingent) regimes
Ω_{BS}	Set of primary substation nodes ($\Omega_{BS} \in \Omega_b$)
Ω_l'	Ω_l without branches candidates for CB/recl. & switches
$\Omega_{CB-RECnew}$	Set of branches candidates for new CB/recloser
$\Omega_{SWITnew}$	Set of branches candidates for new switch

Variables

cr_{ij}^{typ}	Binary construction variables associated with new circuit types in branch ij
nz_{ij}^{typ}	Binary construction variables associated with new switchgear types in branch ij
ce_{ij}	Binary variable for using existing circuit in branch ij
m_i^k	Binary variable denoting circuit connection to primary substation at node i in regime k
P_{Gi}^k	Active power from primary sub at node i in regime k
Q_{Gi}^k	Reactive power from primary sub at node i in regime k
ce'_{ij}	Binary variable equal ce_{ij} for existing branch; 0 otherwise
oc_{ij}^k	Binary operation variable for branch ij in regime k ; equal 1 for closed branches and 0 for open branches
ez_{ij}	Binary variable for using existing switch in branch ij
ez'_{ij}	Binary variable equal ez_{ij} for existing branch; 0 otherwise
V_i^k, θ_i^k	Voltage magnitude and angle at node i in regime k
θ_{ij}^k	Volatage angle difference in branch ij in regime k
f_{ij}^k	Fictitious power flow in branch ij , regime k due to DG
S_{ij}^k	Fictitious customer flow in branch ij in regime k
$ny_{ij}^{typ}, ocy_{ij}^k$	Construction & operation variable for new CB/recl

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$nx_{ij}^{typ}, ocx_{ij}^k$	Construction & operation variable for new switch in ij	$\underline{V}^k, \overline{V}^k$	Minimum and maximum voltage limits in regime k
PLC_i^k	Active power load curtailment at node i in regime k	N_{SW}, N_{DG}	Number of new switches and DG units-DG nodes
Functions		N_{NOP}	Maximum number of NOPs used during outages
$COST_{cct}(\cdot)$	Capital cost of reinforcing and constructing branches	N_{BS}	Maximum number of subs used during outages
$COST_{SW}(\cdot)$	Capital cost of constructing new switchgear	K_i	Fictitious load equal 1 at DG node i ; 0 otherwise
$COST_{dec}(\cdot)$	Cost of circuit decommissioning	q	Number of branches connecting a DG to the network
$P_{ij}^k(\cdot), Q_{ij}^k(\cdot)$	Active & reactive power flows in branch ij in regime k	n_b^k	Total number of nodes in regime k
$I_{ij}^k(\cdot)$	Current magnitude in branch ij in regime k	S_i, N_S	Number of customers at node i and total number
$I_{rij}^k(\cdot), I_{mij}^k(\cdot)$	Real and imaginary part of branch ij current, regime k	T	Length of the reconfiguration period in h and total number
REV_{qos}	Utility revenue under QoS incentive regime	P_k	Probability of operation regime k
$\Delta CI(\cdot)$	SAIFI improvement due to a QoS action (operation stage)	CO_{sub}	Substation annual operation cost in £/(kVAh · yr)
$\Delta CML(\cdot)$	SAIDI improvement due to a QoS action (operation stage)	CM_{sub}	Substation annual maintenance cost in £/(kVA · yr)
$CO\&M_{sub}(\cdot)$	Substation O&M cost in £/reconfiguration period	CM_{Sline}	Line specific annual maintenance cost in £/(km · yr)
$CM_{line}(\cdot)$	Line maintenance cost in £/reconfig. period	CM_{CB-REC}	CB-recloser annual maintenance cost in £/yr
$C_{loss}(\cdot)$	Cost of line losses in £/reconfiguration period	CM_{FUSE}, CM_{SWIT}	Fuse & switch annual maintenance cost in £/yr
$CSW(\cdot)$	Cost of switching in £/reconfiguration period	Ψ_{xxij}	= 0 no switchgear in ij ; =1 with switchgear in ij
$C_{inter}(\cdot)$	Interruption cost in £/reconfiguration period	CS_{loss}	= ny_{ij}^{typ} with new CB/recl; = nx_{ij}^{typ} with new switch
Parameters		Llf	Cost of losses in £/kWh
k_t^{NPV}	NPV factor for investments in time period t	CSW	Line loss factor in the reconfiguration period
$C_{ij}^{br,typ}$	Branch ij capital cost per km for new circuit types	oc_{ij}^0	Cost of a switching – existing switchgear in £/No.Op
$c_{ij}^{ter,typ}$	Branch ij terminal cost for new circuit types	$CCB-RECL$	Initial status of existing switchgear in branch ij
len_{ij}	Length of branch ij	$CSWIT$	Cost of a switching – new CB/recloser in £/No.Op.
dr	Discount rate	ocy_{ij}^0, ocx_{ij}^0	Cost of a switching – new switch in £/No.Op.
$c_{ij}^{SW,typ}$	Costs of new types of switchgear constructed in branch ij	$VOLL_i(\cdot)$	Initial status of new CB/recloser & switch, branch ij
c_{ij}^{dc}	Decommissioning cost per unit length in branch ij	Lfi	Value of lost load at node i
P_{DGi}^k, Q_{DGi}^k	Active and reactive generation at node i , regime k		Load factor at node i in the reconfiguration period
P_{Di}^k, Q_{Di}^k	Active and reactive demand at node i in regime k		
x_{ij}^k	Fault parameter for branch ij : 0 for outage; 1 otherwise		
g_{ij}, b_{ij}, r_{ij}	Conductance, susceptance & resistance of branch ij		
$(\overline{I}_{ij}^k)^{ex}$	Ratings of existing circuit in branch ij , regime k		
$(\overline{I}_{ij}^k)^{new,typ}$	Ratings of new circuit types in branch ij , regime k		
$(\overline{I}_{ij}^k)^{SWex}$	Thermal capacity of existing switchgear in branch ij		
$(\overline{I}_{ij}^k)^{SWnew,typ}$	Thermal capacity of new switchgear types in ij		

I. INTRODUCTION

MODERN distribution networks are characterized by connection of new low carbon technologies (LCTs) and application of ‘smart’ solutions which generally lead to reinforcement deferral and capacity margin reductions. On the other hand, distribution network operators (DNOs) are faced with challenging requirements imposed by the Regulator. Develop-

ment of distribution networks has to follow general planning standards [1], whilst considering multiple operating regimes within the *investment problem* due to new generation and load types. Next, DNOs have to respond to regulatory financial limitations, which are defined under the Information Quality Incentive (IQI) mechanism [2]. More specifically, the Regulator defines benchmark capital allowances and puts 75% weight on it, whilst DNO's own cost assessments are weighted 25% [3]; this can lead to restrictions on capital investments. Changes in the quality-of-supply (QoS) concepts and modifications of the operational practice may be required due to regulatory QoS and losses incentive regimes. All these aspects need to be addressed in modern network planning.

The most recent, comprehensive reviews of the approaches for distribution network planning can be found in [4]–[6]. Ref. [4] presents classifications and the most frequent mathematical formulations, whilst distribution planning with DG integration has been elaborated in [5], [6]. The objective function is usually minimization of one or more cost terms related to capital investments, energy losses, reliability and operation & maintenance, whilst constraints on thermal and voltage limits, full connectivity and radial configuration, as well as individual equipment capabilities define the feasible solution space [4]. An optimal solution is often derived from a multi-objective formulation [7], [9]–[10]; alternatively, a set of potential solutions and the preferred solution can be obtained from the Pareto set [11]–[13]. Planning of distribution networks can be done either within a single period (static planning) [12] or over a longer, multi-period interval (dynamic planning) [14], [15]. Multi-stage models are often defined as a single optimization model that generates an optimal solution over the entire planning period. Although the best ‘complete’ solution is at the hand, this approach is not appropriate in the real-life planning for several reasons: a) Engineering solutions developed by design engineers cannot be incorporated; b) Different non-technical constraints related to construction capabilities, social and environmental effects, etc. cannot be considered; c) It is not possible to verify some aspects that are not modelled; for example, fault levels where new DG units are connected; d) Transition from one year to another is often characterized by technical constraints which need to be modelled and which add to the modelling complexity; e) Modelling of uncertainties can be very hard (or even impossible), because it may be necessary to introduce new uncertainties later in the planning period; f) It may not be possible to solve optimization problem for real-life networks due to the problem complexity; g) Multi-stage models are not used in the real life. Solution of the optimal planning models can use heuristic approaches [16], [17], or more traditional optimization methods [18]–[22].

The objective function of distribution network planning with optimal DG integration can be single- or multi-objective [5], [6]. The most frequently used single objectives are minimization of power or energy losses [23]–[27], minimization of various mixes of operational, investment and reliability costs [20], [21], [28], [29], maximization of DG capacity [30]–[33], as well as maximization of profit or benefit/cost ratio [34]–[36]. Multi-objective functions are usually weighted sums of single objectives, or multi-objective formulation with several often contrast-

ing single objectives [37]–[45]. The proposed models are solved using analytical techniques, traditional optimization techniques, exhaustive search, (meta)heuristic methods, or probabilistic approaches [5], [6]. The latest developments also consider DG control [46] and optimization of DG and electric vehicles connected to the distribution network [47]–[49].

Literature review has shown that network investment optimization models with explicit modelling of security constraints (i.e. network outages included in the model) have only been developed for transmission networks operated as meshes [50]–[52]. The authors are not aware of any paper that solves investment optimization of *radially operated* distribution networks with security constraints, in which switchgear statuses need to be determined for each outage. Complexity of the model is increased by the requirement for radial operation in each outaged regime, which is achieved by radiality constraints and additional modelling when DG units are connected [53]. Next, the optimal *expansion* planning models still do not consider multiple operating regimes and single investment and operation optimization models do not support real-life practice. In that respect, a comprehensive model that integrates all major components of the today's real-life distribution planning within DNOs is still lacking.

In recent years, several comprehensive methodologies that optimize investment, operational, outage, quality-of-supply costs and/or financial benefits in a ‘*single step*’ have been proposed [9], [10], [12], [14], [15], [18], [19], [22]. However, practical applicability of these models in real-life can be challenged because they do not support business structure in utilities. It is for this reason that a methodology for integrated planning of distribution networks based on the UK utility practice is developed in this paper. The proposed approach is based on static models, decision tree concept [54] that can be applied in deterministic or probabilistic way, and the two-stage optimization: a) Investment problem; and b) Operation problem. This paper focuses on development of *investment* optimization models. Explicit modelling of all (N-1) contingencies on the studied MV circuit is modelled in conjunction with multiple operating regimes. During restoration of supplies in system emergencies, new switching devices can be constructed to avoid further reinforcements. To limit the area affected by a fault, utilities often apply specific rules related to the number of neighboring feeders and NOPs involved in restoration; some of rules are included in the model. Next, optimal connection of new DG units is based on the assumption they are owned by third parties and their location is known. All these features led to the development of two non-linear mixed integer investment models. The first ‘full’ model enables construction of new switching devices in the design stage and it is appropriate for rural and mixed networks. The second model does not have this feature and is appropriate for the UK urban networks.

The main contributions of the paper are:

- 1) A flexible global methodology for distribution planning based on the UK real life practice is proposed.
- 2) Uncertainties in the distribution planning are modelled with the aid of a probabilistic decision tree.
- 3) Security constraints and multiple operating regimes are explicitly modelled for radially operated networks.

- 4) Company-specific policies on restoration of supplies are included in the optimization model.
- 5) Construction of new switchgear along with network reinforcements is envisaged to reduce total investment costs. Decommissioning of *branches* is also proposed.
- 6) Limitations on the number of connected customers to a MV feeder are considered in the investment model.
- 7) Operation problem is solved in two steps: a) QoS optimization; and b) Operational cost minimization.

The paper is organized as follows: overall problem formulation is presented in Section II, ‘full’ investment model in Section III, simplified investment model in Section IV, model solution in Section V, outlines of the operational model in Section VI, test networks/results in Section VII, comparative studies in Section VIII and conclusions in Section IX.

II. OVERALL PROBLEM FORMULATION

Outlines of the UK distribution planning and overall problem formulation are briefly explained below.

A. Real-Life Planning of Medium Voltage Networks

Planning of medium voltage distribution networks in the UK utilities is done in four independent stages: a) Replacement planning; b) Reinforcement planning; c) QoS planning; and d) Operations planning. Network replacements are determined by analyzing asset groups and individual assets without considering the entire network; the decision is based either on asset health indices, or failure history, or asset age.

‘General reinforcement’ planning considers increase in general load, approved connections of new DG units and compliance with the national planning standards [1]. This activity is typically done on a yearly basis and replacements driven by network constraints are determined in this stage. Studies of connection applications for both new load and generation are done in parallel on a daily basis. The next activity is QoS studies which are triggered by the regulatory incentive regime and fault statistics (i.e. poor performance). The main goal is to improve *SAIFI (CI)* and *SAIDI (CML)* reliability indices by automating existing switchgear and possibly installing new switchgear. This stage is done in the UK separately from the replacement and reinforcement planning by using bespoke software [55]. Finally, calculation of the optimal network configurations is based on the minimization of total operational costs. In case of non-automated networks it is done once or twice per year, whilst for fully automated networks it can be done every hour, in which case it is called *dynamic* network reconfiguration.

B. Overall Problem Formulation

The overall problem is set in the form of a decision tree [54], where the nodes denote different network configurations in certain time periods and branches transitions between two configurations in consecutive time periods. The entire problem is divided into investment stage, operations stage and calculation of final results, as shown by dashed lines in Fig. 1. The

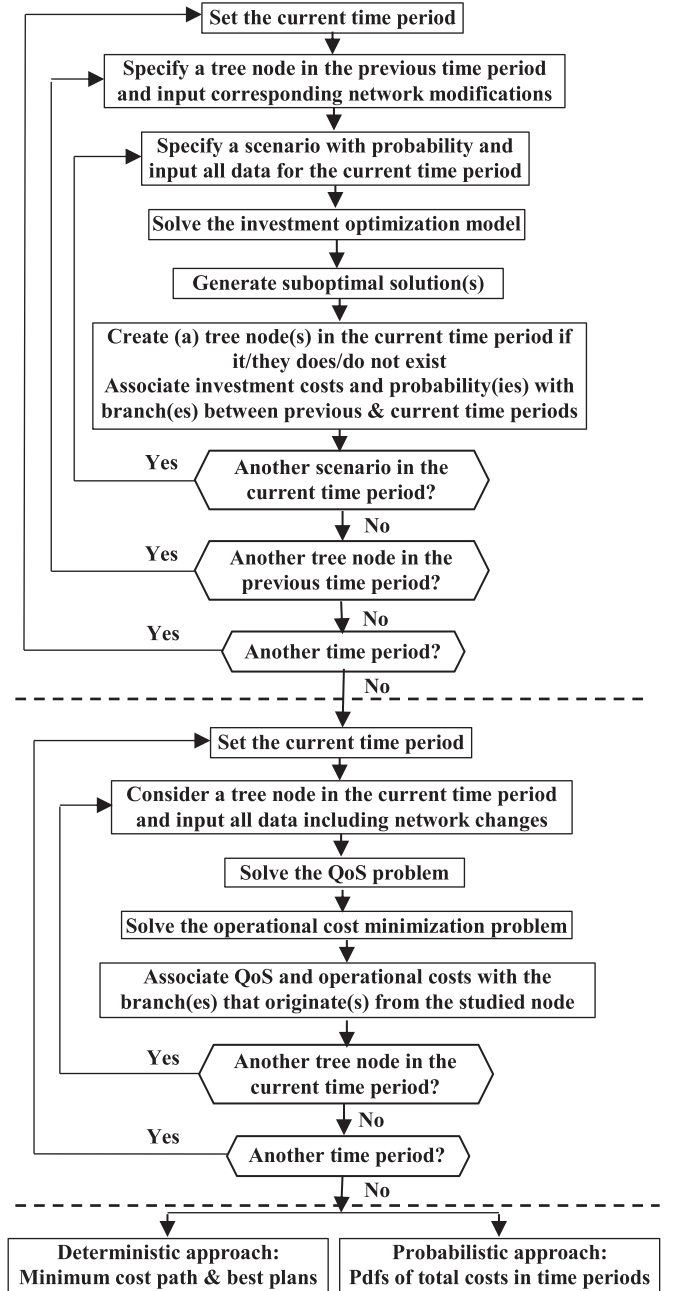


Fig. 1. Global algorithm of the integrated planning methodology.

presented methodology can be used for both deterministic and probabilistic planning.

The nodes of the decision tree are generated in the first stage by solving the investment model that gives capital costs. The analysis always starts from a node in the previous time period and then determines transitions (i.e. branches) towards the newly established nodes in the current time period. Different configurations in the current time period are obtained by including suboptimal solutions and/or by considering some of uncertainties. Discounted capital costs are associated with relevant tree branches. Analysis of the current time period is completed when all scenarios for a node in the previous period are exhausted and when all ‘previous nodes’ are processed. When the analysis of

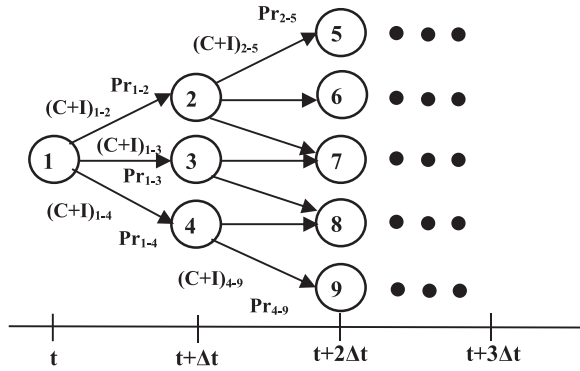


Fig. 2. Illustrative example of a decision tree.

the entire planning period is completed and the decision tree is established, it is proceeded with the solution of the operation problem for each node in each time period (second part below dashed line in Fig. 1). The currently applied approach is done in two steps in line with the UK industry practice. The QoS model determines optimal existing switchgear to be automated and possibly new switchgear to be installed. Minimization of operational costs is the second step; they include operation and maintenance (O&M) costs of substations (note that these costs are defined on a yearly basis), maintenance costs of lines, cost of losses and switching, and interruption costs. The QoS and operation costs are associated with the tree branches which originate from the studied node. When the operational stage is completed in the last year of the planning period, final results are calculated; the difference between deterministic and probabilistic approaches is presented below.

a) *Deterministic Approach.*: Deterministic decision tree approach considers discounted investment and operational costs without uncertainties; they are denoted as $(C + I)_{x-y}$ in Fig. 2. The decision tree can be created using one of the following three concepts:

- 1) 'Horizon year planning': solutions are generated for the horizon year first; temporal planning is then initiated from the first year always 'looking into' the horizon year solutions. The method is usually not used in the regulated industry.
- 2) 'Regulatory year planning': concept very similar to the horizon year planning but applied to the end-year of the regulatory period. It is usually used for the DNO submission to the Regulator for the approval of capital investments.
- 3) 'Minimum cost planning': refers to the planning starting from the first year and looking into a couple of years ahead. It is often driven by the regulatory requirement to provide minimum cost plans at all time periods.

Suboptimal investment solutions can be generated by some optimization techniques. However in real-life, these solutions are developed by design engineers and possibly by looking into the regulatory year development plans. A minimum path algorithm or dynamic programming is used to find the optimum of the overall multi-stage problem (i.e. 'optimal path') in terms of the total minimum costs [54]. In real life, optimal path is of-

ten determined after the first, investment stage, and operational planning is then done only for optimal development plans on the optimal path.

b) *Probabilistic Approach.* This paper proposes a probabilistic decision tree approach to model uncertainties. In this way, design engineers will be able to understand which network modifications are consequences of specified uncertainties. Probabilities of each scenario Pr_{x-y} are associated with each branch between two configurations; for example $Pr_{1-2} + Pr_{1-3} + Pr_{1-4} = 1.0$ in Fig. 2. Probability of each network configuration is equal to the sum of probabilities of all paths that lead to that tree node; for example, probability of node 7 in Fig. 2 is $Pr_7 = Pr_{1-2}Pr_{2-7} + Pr_{1-3}Pr_{3-7}$. The total 'cost of each node' is equal to the path cost in case of a single path, or to the sum of weighted costs of all paths that lead to it. In the latter case, costs of different paths are similar because difference comes from discounting the capital costs of the same assets in different periods. The end result is a discrete probability distribution function (pdf) of the total cost in each time interval studied. A criterion for the 'best' development plan in each period needs to be specified. This can be, for example, plan with the highest probability, a plan between the highest probability and highest cost plan, or even aggregated plans which are similar and the cost difference is small.

The probabilistic approach from Fig. 2 can be extended with suboptimal solutions and relevant uncertainties.

III. 'FULL' INVESTMENT MODEL

The developed investment model gives optimal MV network reinforcements, new switchgear and possibly circuit decommissioning. The model also considers DG units that are owned by third parties on pre-specified locations and determines their optimal connection to the existing network.

The 'full' investment model is deemed appropriate for rural and mixed networks in which new switchgear can be constructed and overhead lines decommissioned. It contains binary 'construction' and 'operation' decision variables; the difference is that a single construction variable is applicable to all studied operating regimes and contingencies, whilst an operation decision variable is associated with each regime and contingency. Details of the model are presented below.

A. Objective Function

In our approach, it was assumed that a capital investment has materialized at the end of the time period when it is required; another approach, such as uniform series of annual payments, can be incorporated in the overall formulation. The net present value (NPV) of each investment in each time period is therefore calculated. It is proposed to include three categories of capital costs in the model: a) Reinforcement/replacement of existing and construction of new circuits on new corridors and for DG connections; b) Construction of new switchgear; and c) Decommissioning of existing circuits.

The NPV of investment costs associated with reinforcement/replacement of existing branches and construction of cir-

cuits in new corridors and for DG connection is:

$$\text{COST}_{cct}(t) = k_t^{\text{NPV}} \sum_{ij \in \Omega_l} \sum_{typ} \left(c_{ij}^{br,typ} \cdot \text{len}_{ij} + c_{ij}^{ter,typ} \right) \cdot cr_{ij}^{typ} \quad (1)$$

where NPV factor is $k_t^{\text{NPV}} = (1 + dr)^{-t}$. Formulation (1) is applicable to any number of new circuit sizes *typ*; we have however used a small number of new types (up to three). This approach is justified by the current practice in DNOs whereby a small number of new asset types is constructed on each voltage level to harmonize asset registry and simplify O&M. Costs (1) can be simply extended with the investment cost for shunt compensation expressed through costs in £/MVar, rating in MVar and decision variable; shunt compensation is not used because it is not in line with the UK standards [1].

Replacement of existing switchgear and construction of switchgear in new branches is defined in a similar way:

$$\text{COST}_{SW}(t) = k_t^{\text{NPV}} \cdot \sum_{ij \in \Omega_l} \sum_{typ} c_{ij}^{\text{SW},typ} \cdot nz_{ij}^{typ} \quad (2)$$

where up to three types of new switchgear are used in the model. Finally, it is proposed to include the NPV cost of decommissioning of *branches* with existing circuits:

$$\text{COST}_{dec}(t) = k_t^{\text{NPV}} \sum_{ij \in \Omega_{lex}} (c_{ij}^{\text{dc}} \text{len}_{ij}) \left[1 - \left(ce_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \right] \quad (3)$$

where individual branch cost is non-zero if all decision variables are zero (i.e. for decommissioned branch). We have used cost proportional to the circuit length, which is appropriate for overhead lines; in case of underground cable decommissioning cost is usually not dependent on circuit length because the cable terminals are only disconnected.

Total cost is the sum of COST_{cct} , COST_{SW} and COST_{dec} . Note that the O&M costs are specified in the operation model because they are defined on an annual basis.

B. Constraints

Constraints of the ‘full’ investment model are classified into three groups: a) Power flow; b) Logical; and c) Other constraints. They are presented by groups so defined.

Power flow constraints are given by relations (4)–(13):

$$m_i^k \cdot P_{Gi}^k + P_{DGi}^k - P_{Di}^k - \sum_{ij \in \Omega_l} \left(ce_{ij}' + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \cdot P_{ij}^k(\cdot) = 0 \quad \{\forall i \in \Omega_b\} \quad \{\forall k \in \Omega_k\} \quad (4)$$

$$m_i^k \cdot Q_{Gi}^k + Q_{DGi}^k - Q_{Di}^k - \sum_{ij \in \Omega_l} \left(ce_{ij}' + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \cdot Q_{ij}^k(\cdot) = 0 \quad \{\forall i \in \Omega_b\} \quad \{\forall k \in \Omega_k\} \quad (5)$$

$$P_{ij}^k(\cdot) = (V_i^k)^2 g_{ij} - V_i^k V_j^k (g_{ij} \cos \theta_{ij}^k + b_{ij} \sin \theta_{ij}^k) \quad (6)$$

$$Q_{ij}^k(\cdot) = -(V_i^k)^2 b_{ij} + V_i^k V_j^k (b_{ij} \cos \theta_{ij}^k - g_{ij} \sin \theta_{ij}^k) \quad (7)$$

$$\left(ce_{ij}' + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \cdot I_{ij}^k(\cdot)^2 \leq \left(ce_{ij}' \cdot (\bar{I}_{ij}^k)^{ex} + \sum_{typ} cr_{ij}^{typ} \cdot (\bar{I}_{ij}^k)^{\text{new},typ} \right)^2 \quad \{\forall ij \in \Omega_l\} \quad (8)$$

$$\left(ce_{ij}' + \sum_{typ} cr_{ij}^{typ} \right) \left(ez_{ij}' + \sum_{typ} nz_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \cdot I_{ij}^k(\cdot)^2 \leq \left(ez_{ij}' \cdot (\bar{I}_{ij}^k)^{\text{SWex}} + \sum_{typ} nz_{ij}^{typ} \cdot (\bar{I}_{ij}^k)^{\text{SWnew},typ} \right)^2 \quad \{\forall ij \in \Omega_l\} \quad (9)$$

$$I_{ij}^k(\cdot) = [I_{rij}^k(\cdot)^2 + I_{mij}^k(\cdot)^2]^{0.5} \quad \{\forall ij \in \Omega_l\} \quad (10)$$

$$I_{rij}^k(\cdot) = g_{ij} (V_i^k \cos \theta_{ij}^k - V_j^k \cos \theta_{ij}^k) - b_{ij} (V_i^k \sin \theta_{ij}^k - V_j^k \sin \theta_{ij}^k) \quad \{\forall ij \in \Omega_l\} \quad (11)$$

$$I_{mij}^k(\cdot) = g_{ij} (V_i^k \sin \theta_{ij}^k - V_j^k \sin \theta_{ij}^k) + b_{ij} (V_i^k \cos \theta_{ij}^k - V_j^k \cos \theta_{ij}^k) \quad \{\forall ij \in \Omega_l\} \quad (12)$$

$$V_-^k \leq V_i^k \leq \bar{V}^k \quad \{\forall i \in \Omega_b\} \quad (13)$$

Full AC load-flow nodal active and reactive power balance (4) and (5) are set for each network configuration and operating regime, which are defined using a single superscript *k*. In our approach, *k* = 0 denotes winter peak – intact network, *k* = 1, ..., *N* winter peak – single contingencies, *k* = *N* + 1 summer minimum – intact network, and *k* = *N* + 2, *N* + 3, ... summer minimum – single outages. The first term in (4) and (5) is applicable to the sending node of the first section of feeders; for intact network $m_i^k = 1$ for the studied feeder, whilst $m_i^k = 0$ or 1 for all backfeeding feeders in case of contingencies. The first term in brackets under summation considers whether a circuit exists or not in branch *i* – *j*; a new variable ce_{ij}' equal to ce_{ij} for existing branches and 0 for all new corridors is introduced to simplify presentation. Branch operation status is defined by binary operation variable oc_{ij}^k , whilst ‘fault parameter’ x_{ij}^k had to be introduced to model single outages because $oc_{ij}^k = 0$ could not be used due to contradiction with logical constraints (17). This parameter is set to zero for an outaged branch and it is equal to unity in all other cases. Finally, branch active and reactive power flows $P_{ij}^k(\cdot)$ and $Q_{ij}^k(\cdot)$ are given by (6) and (7); they are functions of voltage magnitudes and angles at terminal nodes [53].

Circuit thermal constraints (8) are modelled with the aid of branch currents $I_{ij}^k(\cdot)$ that are functions of terminal voltages and angles (10)–(12). Function $I_{ij}^k(\cdot)$ is non-zero even where the circuit is decommissioned or the switch is open, hence the same multipliers as in (4) and (5) are used on the left-hand side of (8). Circuit ratings on the right-hand side are dependent on regime *k*, which is particularly important for overhead lines (i.e. winter vs

summer). Thermal constraints of switchgear (9) are expressed in a similar way recognizing that switchgear must exist in the considered branch ij . Limits on nodal voltage magnitudes (13) can be different for intact network and contingencies, which is often the case in real-life planning of rural areas.

Logical constraints are defined by relations (14)–(20):

$$ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \leq 1 \quad \{\forall ij \in \Omega_l\} \quad (14)$$

$$ez'_{ij} + \sum_{typ} nz_{ij}^{typ} \leq 1 \quad \{\forall ij \in \Omega_{SW}\} \quad (15)$$

$$\left[1 - \left(ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \right] \cdot \left(ez'_{ij} + \sum_{typ} nz_{ij}^{typ} \right) = 0 \quad \{\forall ij \in \Omega_l\} \quad (16)$$

$$(1 - oc_{ij}^k) \cdot \left[1 - \left(ez'_{ij} + \sum_{typ} nz_{ij}^{typ} \right) \right] = 0 \quad \{\forall ij \in \Omega_{SW}\} \quad (17)$$

$$\sum_{ij \in \Omega_{SWnew}} \sum_{typ} nz_{ij}^{typ} \leq N_{SW} \quad (18)$$

$$\sum_{\Omega_{NOP}} oc_{ij}^k \leq N_{NOP} \quad \{\forall k \in \Omega_k^{abn}\} \quad (19)$$

$$\sum_{\Omega_{BS}} m_i^k \leq N_{BS} \quad \{\forall k \in \Omega_k^{abn}\} \quad (20)$$

Inequality (14) shows that branch ij can be either in use, or decommissioned. Existence of switchgear in branch ij is defined in a similar way through inequalities (15). A new variable, ez'_{ij} , that is equal to ez_{ij} for existing branches and 0 for new ones, is used to simplify presentation. Next, replacement or construction of new switchgear has to be linked to the presence of circuitry in existing and new branches, which is modelled with relations (16). Finally, operational status of switchgear expressed through oc_{ij}^k needs to be linked to the presence of switchgear in each branch – constraints (17). If there is no switchgear in branch ij the branch is closed, i.e. $oc_{ij}^k = 1$.

DNOs sometimes impose restrictions on the number of switching devices that can be installed on a circuit – relation (18). Besides, company-specific restoration policy needs to be defined in case of circuit outages, because the ‘whole’ network around the faulty feeder cannot be used in real-life. We have used simple rules (19) and (20) applied in the UK utilities; they specify maximum number of back-feeding feeders (20) and normally open switches between the faulty and back-feeding feeders (19). It is possible to define other restoration policies, such as minimization of the number of switching operations or the number of back-feeding feeders, by extending the objective function.

The last group of constraints is given by relations (21)–(27):

$$\sum_{ij \in \Omega_l} \left(ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \cdot f_{ij}^k = K_i \quad \{\forall i \in \Omega_b\} \quad (21)$$

$$|f_{ij}^k| \leq N_{DG} \cdot \left(ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot x_{ij}^k \quad \{\forall ij \in \Omega_l\} \quad (22)$$

$$\sum_{ij \in \Omega_{Con}} \sum_{typ} cr_{ij}^{typ} = q \quad (23)$$

$$\sum_{ij \in \Omega_l} (ce'_{ij} + \sum_{typ} cr_{ij}^{typ}) \cdot oc_{ij}^k = n_b^k - 1 \quad \{\forall k \in \Omega_k^{nor}\} \quad (24)$$

$$\sum_{ij \in \Omega_l} (ce'_{ij} + \sum_{typ} cr_{ij}^{typ}) \cdot oc_{ij}^k \cdot x_{ij}^k = n_b^k - \sum_{\Omega_{BS}} m_i^k \{k \in \Omega_k^{abn}\} \quad (25)$$

$$\sum_{ij \in \Omega_l} \left(ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \cdot S_{ij}^k = S_i \quad \{\forall i \in \Omega_b\} \quad k = 0 \quad (26)$$

$$|S_{ij}^k| \leq N_s \cdot \left(ce'_{ij} + \sum_{typ} cr_{ij}^{typ} \right) \cdot oc_{ij}^k \quad \{\forall ij \in \Omega_l\} \quad (27)$$

When DG units are connected to a distribution network, there is a possibility that the optimal solution contains (an) isolated island(s) in which all loads are supplied by DG units. To prevent this situation, concept of ‘fictitious power flows’ has been introduced in [53]; its main idea is to define fictitious loads at nodes with DG units, which would generate fictitious power flows and the network islanding is prohibited. Fictitious power balance (21) are specified for each node, with fictitious load $K_i = 1$ for DG nodes and $K_i = 0$ for nodes without DG; fictitious power flows are also limited via constraints (22). Note that fictitious power flows f_{ij}^k are in no way connected to real power flows $P_{ij}^k(\cdot)$ and $Q_{ij}^k(\cdot)$ and they are in fact a part of requirements for radial configuration of the entire feeders in each regime k . Relations (21) and (22) are therefore expressed through construction and operation decision variables. Next, it was assumed that DG units are owned by developers, in which case DG locations are known and they need to be connected to the network in the best way. Then, constraints on the number of DG connecting branches (23) need to be specified; here, parameter $q = 1$ is for ‘non-firm’ connection (i.e. DG is teed off) and $q = 2$ for ‘firm’ connection (i.e. DG is looped in). Radial configuration of the optimal solution has to be ensured for each regime k ; this is expressed through a set of constraints related to intact network (24), as well as to contingent configurations (25). Finally, constraints (26) and (27) address the allowed number of connected customers, which is usually specified in DNO network design policy documents [56]. These constraints are ‘similar’ to fictitious power flow (21) and (22) and need to be specified within the investment stage [56].

Variables of the model (1) to (27) are binary decision variables ce_{ij} , cr_{ij}^{typ} , ez_{ij} , nz_{ij}^{typ} and oc_{ij}^k , nodal voltage magnitudes and angles V_i^k and θ_i^k in all studied regimes k , as well as fictitious power flows f_{ij}^k and customer flows S_{ij}^k .

IV. 'SIMPLIFIED' INVESTMENT MODEL

The essential assumptions are: a) Decommissioning of branches is not permitted; b) New switchgear is not installed because each branch contains (a) switch(es); and c) Two new asset types can be constructed. The model is appropriate for the UK urban (cable) networks, where ground mounted distribution substations are connected on MV via RMUs, cable decommissioning is very infrequent and capital costs are almost constant for different cable sizes.

Objective function is given by (1); constraints are:

$$m_i^k \cdot P_{Gi}^k + P_{DGi}^k - P_{Di}^k - \sum_{ij \in \Omega_{lex}} [oc_{ij}^k \cdot P_{ij}^k(\cdot)] - \sum_{ij \in \Omega_{lnew}} (cr_{ij}^1 + cr_{ij}^2) \cdot oc_{ij}^k \cdot P_{ij}^k(\cdot) = 0 \quad \{\forall i \in \Omega_b\} \quad \forall k \in \Omega_k \quad (28)$$

$$m_i^k \cdot Q_{Gi}^k + Q_{DGi}^k - Q_{Di}^k - \sum_{ij \in \Omega_{lex}} [oc_{ij}^k \cdot Q_{ij}^k(\cdot)] - \sum_{ij \in \Omega_{lnew}} (cr_{ij}^1 + cr_{ij}^2) \cdot oc_{ij}^k \cdot Q_{ij}^k(\cdot) = 0 \quad \{\forall i \in \Omega_b\} \quad \forall k \in \Omega_k \quad (29)$$

$$oc_{ij}^k \cdot I_{ij}^k(\cdot)^2 \leq \left(ce_{ij} \cdot (\bar{I}_{ij}^k)^{ex} + cr_{ij}^1 \cdot (\bar{I}_{ij}^k)^{new,1} + cr_{ij}^2 \cdot (\bar{I}_{ij}^k)^{new,2} \right) \quad \{\forall ij \in \Omega_{lex}\} \quad (30)$$

$$(cr_{ij}^1 + cr_{ij}^2) \cdot oc_{ij}^k \cdot I_{ij}^k(\cdot)^2 \leq \left(cr_{ij}^1 \cdot (\bar{I}_{ij}^k)^{new,1} + cr_{ij}^2 \cdot (\bar{I}_{ij}^k)^{new,2} \right) \quad \{\forall ij \in \Omega_{lnew}\} \quad (31)$$

$$V_-^k \leq V_i^k \leq \bar{V}^k \quad \{\forall i \in \Omega_b\} \quad (32)$$

$$ce_{ij} + cr_{ij}^1 + cr_{ij}^2 = 1 \quad \{\forall ij \in \Omega_l\} \quad (33)$$

$$\sum_{\Omega_{NOP}} oc_{ij}^k \leq N_{NOP} \quad \{\forall k \in \Omega_k^{abn}\} \quad (34)$$

$$\sum_{\Omega_{BS}} m_i^k \leq N_{BS} \quad \{\forall k \in \Omega_k^{abn}\} \quad (35)$$

$$\sum_{ij \in \Omega_{lex}} oc_{ij}^k \cdot f_{ij}^k + \sum_{ij \in \Omega_{lnew}} (cr_{ij}^1 + cr_{ij}^2) \cdot oc_{ij}^k \cdot f_{ij}^k = K_i \quad \{\forall i \in \Omega_b\} \quad (36)$$

$$|f_{ij}^k| \leq N_{DG} \cdot oc_{ij}^k \quad \{\forall ij \in \Omega_{lex}\} \quad (37)$$

$$|f_{ij}^k| \leq N_{DG} \cdot oc_{ij}^k \cdot (cr_{ij}^1 + cr_{ij}^2) \quad \{\forall ij \in \Omega_{lnew}\} \quad (38)$$

$$\sum_{ij \in \Omega_{con}} (cr_{ij}^1 + cr_{ij}^2) = q \quad (39)$$

$$\sum_{ij \in \Omega_{lex}} oc_{ij}^k + \sum_{ij \in \Omega_{lnew}} oc_{ij}^k \cdot (cr_{ij}^1 + cr_{ij}^2) = n_b^k - 1 \quad \{\forall k \in \Omega_k^{nor}\} \quad (40)$$

$$\sum_{ij \in \Omega_{lex}} oc_{ij}^k + \sum_{ij \in \Omega_{lnew}} oc_{ij}^k \cdot (cr_{ij}^1 + cr_{ij}^2) = n_b^k - \sum_{\Omega_{BS}} m_i^k \quad \{k \in \Omega_k^{abn}\} \quad (41)$$

$$\sum_{ij \in \Omega_{lex}} oc_{ij}^k \cdot S_{ij}^k + \sum_{ij \in \Omega_{lnew}} oc_{ij}^k \cdot (cr_{ij}^1 + cr_{ij}^2) \cdot S_{ij}^k = S_i \quad \{\forall i \in \Omega_b\} \quad k = 0 \quad (42)$$

$$|S_{ij}^k| \leq N_s \cdot oc_{ij}^k \quad \{\forall ij \in \Omega_{lex}\} \quad (43)$$

$$|S_{ij}^k| \leq N_s \cdot oc_{ij}^k \cdot (cr_{ij}^1 + cr_{ij}^2) \quad \{\forall ij \in \Omega_{lnew}\} \quad (44)$$

Modelling of existing branches considers operation variable oc_{ij}^k because decommissioning is not envisaged (constraint (33) is equality), whilst both construction variables cr_{ij}^1 and cr_{ij}^2 and operation variable oc_{ij}^k are needed for new branches on new corridors and connecting DG units; this is recognized in nodal power flow (28) and (29), circuit thermal constraints (30) and (31), expressions (36), (37) and (38) related to fictitious power flows, radiality constraints (40) and (41), as well as constraints on the maximum number of connected customers (42), (43) and (44). Voltage constraints (32), restoration constraints (34) and (35), and the constraint on the DG connection policy (39) are unchanged. Branch active and reactive power flows are defined by (6) and (7), whilst branch currents by (10)–(12). Modelling of single outages is done by setting $oc_{ij}^k = 0$ in relevant regimes k .

V. MODEL SOLUTION

The developed MINLP models are solved using the commercial package AIMMS, which utilizes CONOPT as the nonlinear solver and CPLEX as the mixed-integer solver [57]. CONOPT solver makes use of the generalized reduced gradient method [58], whilst CPLEX is based on the branch-and-bound method [59]. An auxiliary optimization model was developed to initialize the actual models (particularly binary variables) and speed up their convergence. The initialization model is based on the 'simplified' investment model (1), (28)–(41) and the following simplifying assumptions: a) First Kirchhoff's Law is only considered in (28); branch flows fl_{ij}^k are used instead of $P_{ij}^k(\cdot)$; b) Reactive power balances (29) and voltage constraints (32) are not required; c) Thermal constraints (30) and (31) are expressed in terms of flows fl_{ij}^k and appropriate MVA ratings scaled by power factors; d) One type of new assets is considered; and e) Constraints on 'customer flows' (42)–(44) are not considered. The most important outages and operating regimes are incorporated in the model, so that 'good' execution times are obtained. All runs are done on a computer with i7 processor, 3.4 GHz, 32 GB RAM.

VI. OUTLINES OF THE OPERATION MODEL

The current approach is to solve the operation model in two stages, which is in line with the UK utility practices (Section II-A): a) QoS planning; and b) Operational cost minimization via (re)configuration. Note that a QoS programme is built (or not) once in a couple of years, whilst (re)configuration of networks is done at least on a yearly basis.

A. Quality-of-Supply Planning

QoS planning is divided into four steps, as illustrated in Fig. 3: a) Pre-processing; b) Switchgear construction; c) Switchgear

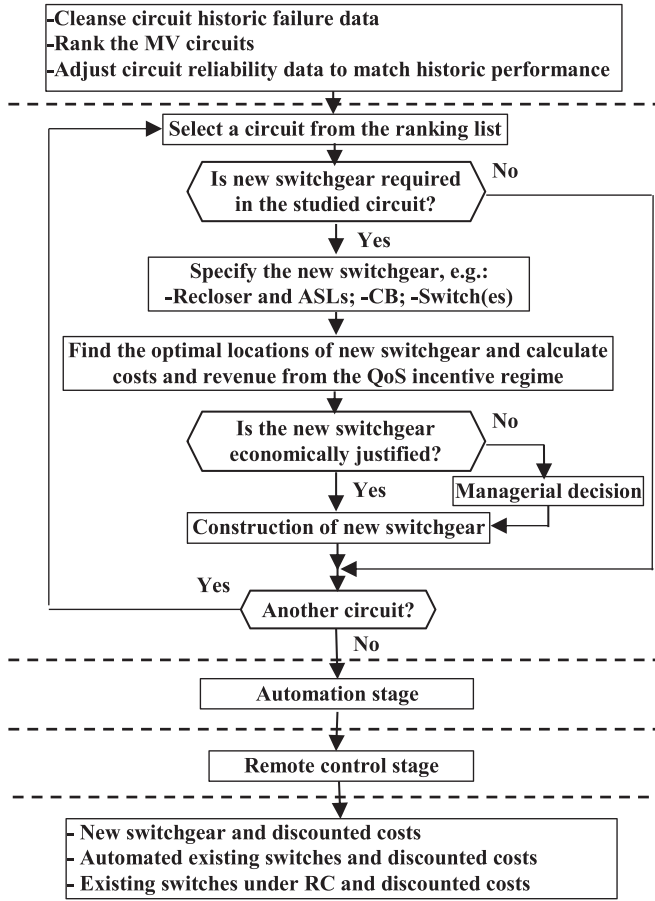


Fig. 3. Global algorithm of the QoS planning.

automation; and d) Remote control of switchgear. Pre-processing starts with cleansing of circuit failure data held in the national database [60]. MV circuits are then ranked starting with the worst performing, and reliability parameters (i.e. failure rates, switching and repair times) are adjusted in the network models to get the true historic performance.

Construction of new switchgear is done in the second stage if automation and remote control are not sufficient. New reclosers with coordinated automatic sectionalizing links (ASLs), switches and sometimes drop-out fuses are installed on rural and mixed circuits, whilst mid-point circuit breaker is sometimes constructed on urban circuits. Assuming the regulatory regime has QoS incentive mechanism, the regulator usually does not provide additional allowance for QoS capital expenditure and QoS investments need to be economically viable. This is based on the comparison of costs with the maximized QoS revenue from the incentive regime, e.g.:

$$REV_{QoS} = \sum_t \Delta CI(t) * CI_{Inc} + \Delta CML(t) * CML_{Inc} \quad (45)$$

where summation goes over the next (or two) regulated period(s). Note that (45) can only be applied where the incentive regime is symmetric with no dead-band, which is the case in the UK. If the investment is not covered by the QoS revenue,

the construction can still go ahead; for example, if it is covered by other more beneficial circuits, or based on social, reputational and other grounds.

Automation and remote control stages are very similar to the construction phase (Fig. 3). Automation of pairs of switches is usually done, starting with NOPs and mid-points. Total *SAIFI* (*CI*) and *SAIDI* (*CML*) benefits (45) are maximized and compared with the cost of automation; similar decision making as before is applied. On the other hand, installation of remote control improves only *SAIDI* so that $\Delta CI = 0$ in (45).

The QoS stage is done with PowerFactory software [61].

B. Minimization of Operational Costs

Network reconfiguration is used to minimize operational costs. Level of network automation determines how often it is done. To be able to study different approaches to network reconfiguration, 'time loop' is taken outside the optimization model. The model is repeatedly solved number of years (e.g. 1 year or 5 years) times number of reconfigurations per year.

The 'optimal' location of new switchgear (if any) was based on the maximization of the QoS revenue (45). However, it is assumed at this stage that their location can be varied in the area close to the previous 'optimal' solution; the final optimal location is found considering a composite objective function.

The objective function consists of five terms: a) Operation and maintenance costs of substations; b) Maintenance costs of lines; c) Cost of variable losses; d) Cost of switching; and e) Interruption costs. They are given by relations (46)–(50):

$$COM_{sub}(t) = k_t^{NPV} (CO_{sub} + CM_{sub}/8760) \cdot T \sum_{k \in \Omega_k} Pr_k \sum_i m_i^k \cdot S_i^{\max} \quad (46)$$

$$CM_{line}(t) = k_t^{NPV} \cdot T \sum_{k \in \Omega_k} Pr_k \cdot \sum_{ij \in \Omega_l} oc_{ij}^k \cdot x_{ij}^k \cdot \left[CMS_{line} \cdot \frac{\text{len}_{ij}}{8760} + CM_{CB-REC} \cdot \frac{\psi_{CB-RECij}}{8760} + CM_{FUSE} \cdot \frac{\psi_{FUSEij}}{8760} + CM_{SWIT} \cdot \frac{\psi_{SWITij}}{8760} \right];$$

$$\psi_{xxxij} = 0 \text{ or } 1 \text{ or } ny_{ij}^{typ} \text{ or } nx_{ij}^{typ} \quad (47)$$

$$C_{loss}(t) = k_t^{NPV} \cdot CS_{loss} \cdot T \sum_{k \in \Omega_k} Pr_k \cdot \left[\sum_{ij \in \Omega_l} oc_{ij}^k \cdot Llf \cdot r_{ij} \cdot I_{ij}^k(\cdot)^2 + \sum_{ij \in \Omega_{CB-RECnew}} \sum_{typ} ny_{ij}^{typ} \cdot oc_{ij}^k \cdot x_{ij}^k \cdot Llf \cdot r_{ij} \cdot I_{ij}^k(\cdot)^2 + \sum_{ij \in \Omega_{SWITnew}} \sum_{typ} nx_{ij}^{typ} \cdot oc_{ij}^k \cdot x_{ij}^k \cdot Llf \cdot r_{ij} \cdot I_{ij}^k(\cdot)^2 \right] \quad (48)$$

$$\begin{aligned}
CSW(t) = & k_t^{NPV} \sum_{k \in \Omega_k} Pr_k \{ C_{SW} \cdot \sum_{ij \in \Omega_i^k} [oc_{ij}^k - oc_{ij}^0]^2 \\
& + C_{CB-RECL} \cdot \sum_{ij \in \Omega_{CB-RECL}} \sum_{typ} ny_{ij}^{typ} \cdot [ocy_{ij}^k - ocy_{ij}^0]^2 \\
& \cdot x_{ij}^k + C_{SWIT} \sum_{ij \in \Omega_{SWIT}} \sum_{typ} nx_{ij}^{typ} [ocx_{ij}^k - ocx_{ij}^0]^2 - x_{ij}^k \}
\end{aligned} \quad (49)$$

$$C_{inter}(t) = k_t^{NPV} T \sum_{k \in \Omega_k} Pr_k \sum_{i \in \Omega_b} VOLL_i(T) \cdot Lf_i \cdot PLC_i^k \quad (50)$$

Substation O&M costs (46) are calculated by multiplying sum of £/kVAh and £/kVA/8760 costs by reconfiguration period length and substation rating for each considered regime k with associated probability Pr_k . Line maintenance costs (47) are expressed in a similar way via specific circuit maintenance costs in £/(km · yr) and maintenance costs of different types of switchgear in £/yr. Quantity ψ_{xxij} is 0 if branch ij does not have switchgear, it is 1 for relevant switchgear in branch ij , it is ny_{ij}^{typ} for new CBs – reclosers and nx_{ij}^{typ} for new switches in branches ij candidates for their installation. Cost of variable losses (48) is specified via line loss factors Lf . Cost of switching (49) is a sum of three terms; it is expressed with the aid of the cost of one switching in £/No.Op and the difference in switchgear statuses in each regime k and the (optimal) intact network operation. Finally, interruption costs (50) are defined via load curtailments at each node and $VOLL$.

Model constraints are ‘similar’ to constraints (4)–(27); the differences are: a) Circuit binary variables ce_{ij}^k and cr_{ij}^{typ} and switchgear variable ez_{ij}^k are not used, b) Construction variable nz_{ij}^{typ} is replaced with two variables ny_{ij}^{typ} and nx_{ij}^{typ} ; c) Load curtailments PLC_i^k are introduced; and d) Constraints (14)–(16), (23), (26) and (27) are not set.

The discounted QoS and operation costs are associated with the decision tree branches that originate from the studied node (Fig. 2). The NPV factor in (46)–(50) is defined for the mid-point of the studied ‘network reconfiguration period’.

VII. TEST NETWORKS AND RESULTS

Three test systems of various scales [62]–[66] were considered to validate the proposed method. Two operating regimes, winter peak and summer minimum, are combined with the intact and contingent network operation. ‘Critical’ branch outages were selected for contingency cases based on severity of these regimes. Summer minimum loads are assumed to be 40% of the winter peaks and all electrical quantities are expressed on a per unit basis. The number of customers at each node is derived by assuming 1.5 kVA of load demand per customer. Other parameters used in the case studies are listed in Table I; reinforcement costs are applicable for the cable ‘excavate and lay’ policy.

A. Case Study I: 33 Bus Test System

The first studied network [62] consists of four 33kV feeders which are supplied from a single primary substation, as shown in

TABLE I
PARAMETERS IN OPTIMIZATION MODELS

Parameter	Value
Discount Rate	5 %
Branch Rein./Construction Cost (£/km)	£44,600/km
Switchgear Construction Cost (£)	£5,000
Branch Decommissioning Cost (£/km)	£1,000/km
Branch Lengths (km)	0.45 km – 2.0 km
Base kVA	33-bus system: 1285 kVA
	69-bus system: 2800 kVA
	119-bus system: 4000 kVA

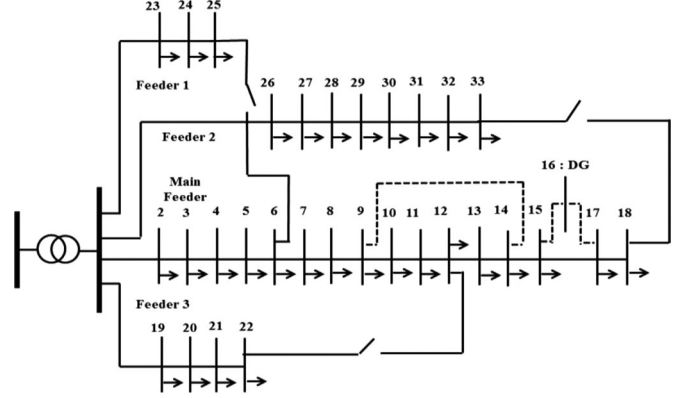


Fig. 4. Single line diagram of the 33-bus test system.

TABLE II
TOTAL INVESTMENT COSTS FOR 33-BUS SYSTEM

Planning Period 1			Planning Period 2		
Scenario	Load Level	Inv. Costs	Scenario	Load Level	Inv. Costs
S1	120%	£ 187 363	S4	120%	£ 175 346
			S5	100%	£ 64 748
			S6	80%	£ 22 567
S2	100%	£ 111 008	S7	120%	£ 195 966
			S8	100%	£ 104 955
			S9	80%	£ 22 567
S3	80%	£ 24 958	S10	120%	£ 216 870
			S11	100%	£ 149 873
			S12	80%	£ 67 486

Fig. 4. The studied feeder is named ‘Main’, ‘Feeder 1’, ‘Feeder 2’ and ‘Feeder 3’ can be used for backfeeding in case of section outages, whilst new corridors and DG connecting branches are marked with dashed lines. Winter peak and summer minimum are combined with intact network, first Section 1-2 and branch 9-10 & 12-13 outages.

Probabilistic decision tree concept (Fig. 2) is applied for the first two time periods in years 5 and 10. The assumed uncertainty is related to the load level, which can be 120%, 100% and 80%, with probabilities 20%, 60% and 20% respectively. This approach gives three tree nodes at the end of period 1 and nine tree nodes at the end of period 2, as shown in Table II by scenarios S1–S12. It is assumed that the load growth between two periods is 25%.

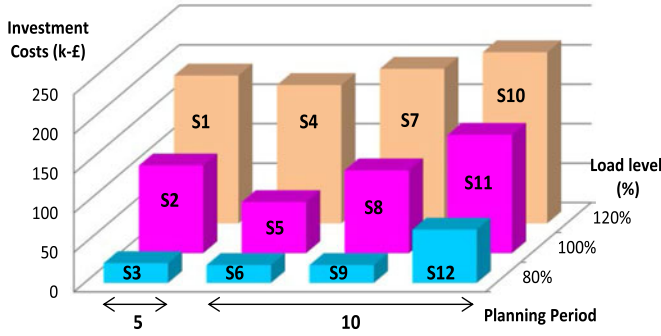


Fig. 5. Plot of investment costs by scenario for 33-bus test system.

Total discounted investment costs required for all scenarios in both periods S1–S12 are given in Table II. Associated probabilities are: $P_{S1} = 0.2$; $P_{S2} = 0.6$; $P_{S3} = 0.2$; $P_{S4} = 0.04$; $P_{S5} = 0.12$; $P_{S6} = 0.04$; $P_{S7} = 0.12$; $P_{S8} = 0.36$; $P_{S9} = 0.12$; $P_{S10} = 0.04$; $P_{S11} = 0.12$; and $P_{S12} = 0.04$. In all cases, highest investment costs are observed for 120% load level, followed by 100% and 80% load levels, because the network is demand dominated and highest power flows are in winter peak regime.

The investment costs associated with each scenario are plotted in Fig. 5. Comparing the investments required for the same load level in planning period 2 (denoted by ‘10’ on x-axis in Fig. 5), it can be concluded that more capital expenditure is required in this period when smaller investments materialized in planning period 1. For example, smallest investments in period 2 was required for S4, followed by S7 and S10 for load level of 120%, as the biggest capital expenditure had been incurred in S1 followed by S2 and S3.

Details of the capital investments for each scenario S1–S12 are listed in Table III. Circuits on new corridors and DG connecting branches, as well as new switchgear are underlined in this table. Other asset interventions denote circuit reinforcements/replacements with bigger conductor sizes, new switchgear with bigger ratings and branch decommissioning. As expected, the highest number of circuit reinforcements was required for backfeeding branches close to the substation. Identical solution was obtained for scenarios S6 and S9 in planning period 2; they are still different decision tree nodes because they originate from scenarios S1 and S2, which have completely different network modifications. Some of the branches that have been reinforced in planning period 1 required another reinforcement (i.e. replacement) in planning period 2. For example, branches 23–24 and 26–27 in S1 and S4, branch 23–24 in S1 and S5, etc. This is the case because the ‘minimum cost planning’ method (Section II-B.a) was adopted in each planning period. If the planning starting from the first year were done looking ahead into the final year solutions (i.e. ‘horizon year’ or regulatory year planning’—Section II-B.a), reinforcements with suitable capacity would have been constructed when the investment is first required, avoiding another reinforcement in the subsequent period; a minimum of 39% total cost reduction would have been obtained compared to the investment costs listed in Table II. Branch decommissioning was a part of the

TABLE III
ASSET CONSTRUCTION & REINFORCEMENT FOR 33-BUS SYSTEM

Scenario	Assets Construction and Reinforcement
S1	Branch: <u>16-17</u> , 19-20, 23-24, 26-27, 27-28, 28-29, 29-30 Switchgear: <u>6-7</u>
S2	Branch: <u>16-17</u> , 23-24, 27-28, 28-29 Switchgear: <u>6-7</u>
S3	Branch: <u>16-17</u>
S4	Branch: <u>9-14</u> , 1-23, 1-26, 2-3, 6-25, 23-24, 24-25, 26-27 Switchgear: 1-2, 1-23, 1-26, 6-25 Branch Decommissioning: 9-10
S5	Branch: 19-20, 20-21, 23-24 Switchgear: 12-22
S6	Branch: 23-24
S7	Branch: <u>9-14</u> , 1-23, 1-26, 2-3, 6-25, 23-24, 24-25, 26-27, 29-30 Switchgear: 1-2, 1-23, 1-26, 6-25 Branch Decommissioning: 12-13
S8	Branch: 19-20, 20-21, 23-24, 26-27, 29-30 Switchgear: 12-22
S9	Branch: 23-24
S10	Branch: 1-23, 1-26, 2-3, 6-25, 23-24, 24-25, 26-27, 27-28, 28-29, 29-30 Switchgear: 1-2, 1-23, 1-26, 6-25
S11	Branch: 19-20, 20-21, 23-24, 26-27, 27-28, 28-29, 29-30 Switchgear: <u>6-7</u> , 12-22
S12	Branch: 23-24, 27-28, 28-29 Switchgear: <u>6-7</u>

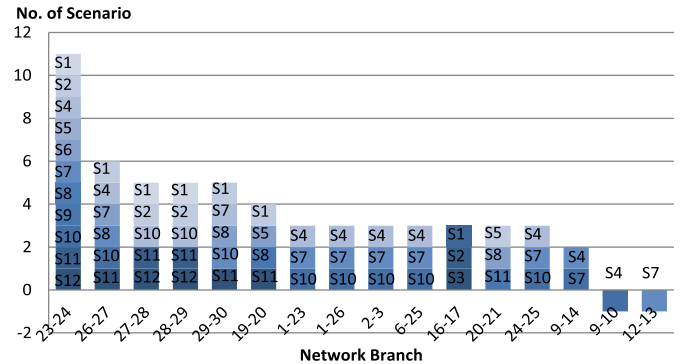


Fig. 6. Branch reinforcements and decommissioning for all scenarios.

minimum cost solution in only two scenarios, i.e. S4 and S7. This can be explained by ‘inflexibility’ of this solution, because once applied it is valid for all studied regimes. On the other hand, more expensive solutions, like installation/replacement of new switchgear, are much more frequently used because of its flexibility to operate in different mode (i.e. open/close) in each studied regime. It can also be seen that decommissioning of branch 9–10 in S4 and branch 12–13 in S7 has resulted in construction of a new branch 9–14. In all scenarios, DG unit was connected to the network via the same, single branch 16–17, because ‘non-firm’ connection was required. Radial operation of the network was achieved in all studied regimes and contingency cases.

Reinforcements and decommissioning of branches are shown in Fig. 6 for all considered scenarios. It can be noticed that branch 23–24 had to be reinforced in all scenarios except S3,

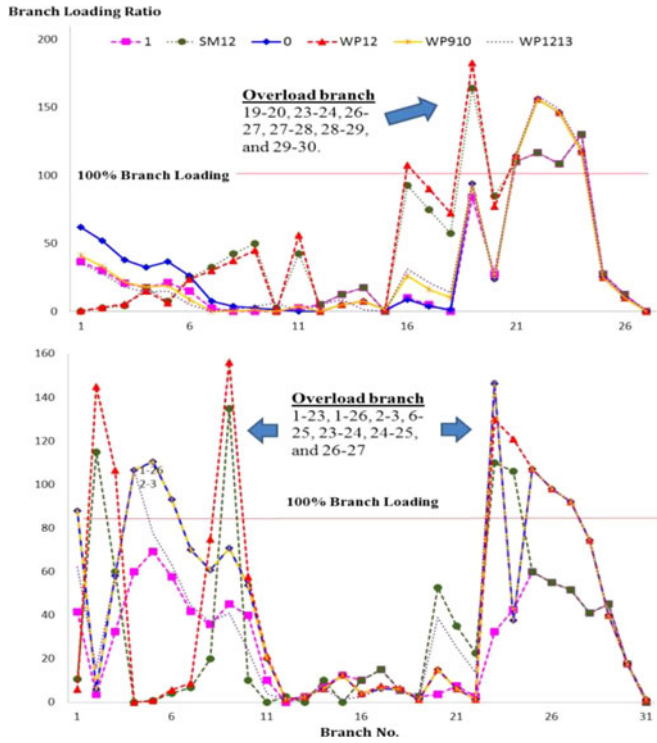


Fig. 7. Ratio of branch currents for all considered regimes in 33-bus test system: Scenario 1 (top), Scenario 4 (bottom).

as ‘Feeder 1’ (Fig. 4) was the preferred backfeeding option during outages; note that thermal rating of branch 1-23 is higher compared to branch 23-24 and thus it needed to be reinforced in 3 scenarios. Circuit decommissioning is marked with ‘downward’ bars at the right-hand side of Fig. 6.

Percentage ratios of branch current flows to their thermal ratings are shown in Fig. 7 for scenarios S1 (upper part) and S4 (lower part) for winter peak (WP) and summer minimum (SM) intact (0 for WP; 1 for SM) and contingency regimes (e.g. WP12 for Section 1-2 outage). Points above the horizontal red line denote branches with loading above their thermal ratings, which required reinforcement as listed in Table III. The highest branch loadings were obtained during winter peak regimes since the network is demand dominated; note that summer ratings were lower than winter ratings. The highest voltage in planning period 2 was 1.018 at node 16 in summer minimum, intact network, whilst the lowest voltage was 0.931 at node 10 in winter peak, first section outage (S4), because resupply was done via branch 6-25, branch 9-10 was decommissioned and node 10 supplied via new branch 9-14. Lower voltage limit was 0.95 for intact network regimes and 0.9 for contingent regimes.

Comparison of the proposed model with established method [17] was possible under the following assumptions: a) DG units are not connected; b) Construction of switchgear is not possible – we have assumed that all branches can be open or closed; and c) There is no circuit decommissioning. We have therefore compared the simplified model (1), (28)–(44) with the model [17]; summary of results is shown in Table IV. The difference in results is the consequence of new branch 9-14 constructed in all considered regimes of the simplified model. When using

TABLE IV
COMPARISON OF THE PROPOSED MODEL WITH METHOD [17]

Branch	Simplified Model	Model [17]	Branch	Simplified Model	Model [17]
9-14	✓	✓	21-22	✓	✓
11-12		✓	26-27	✓	✓
12-22	✓	✓	27-28	✓	✓
19-20	✓	✓	28-29	✓	✓
20-21	✓	✓	29-30	✓	✓
Total Costs		Simplified Model	£ 293 376		£ 327 460
		Model [17]			

TABLE V
COMPUTATIONAL ASPECTS OF ALL CASE STUDIES

Test Systems	33-bus	69-bus	119-bus
Computation Time	338 sec	1089 sec	3710 sec
Memory Size	532.8 Mb	402.8 Mb	2428 Mb

method in [17], construction of branch 9-14 is required for intact regime, while branch 11-12 reinforcement provides cheaper solution for the first section outage, resulting in higher total costs for all regimes considered one at a time. Total non-discounted investment costs show superiority of the proposed method, even when new switchgear construction and branch decommissioning is not used.

The average computation times and memory sizes are shown in Table V. The required computation time depends on combination of many factors, such as the number of locations – candidates for installing new assets, the total number of buses, branches and feeders, as well as the total load level of the overall distribution network. On the other hand, parallel thread options offered through CPLEX solver enable parallel computation to shorten computation time. A nonlinear pre-solver in AIMMS helps to find feasible solution by employing techniques, such as inversion of nonlinear expressions, solving the variables in the pre-triangular part of the matrix, and tightening the bounds of variables based on the linear and nonlinear constraints. The initialization model (Section V) narrows down the search space and speeds up computations.

B. Case Study II: 69 Bus Test System

Another test system that has been studied is the 69 bus system [63], which consists of three 12.66 kV feeders. It was assumed that there are three new corridors 13-21, 27-65 and 50-59 for new circuit construction, as well as two new DG1 connecting branches 62-70 and 63-70 (dashed lines in Fig. 8). DG1 at node 70 needs to be connected in planning period 1, whilst DG2 at node 48 will be added in planning period 2. Our goal is to investigate impact of the number and size of DG units in combination with load levels, indicating that summer minimum is very important. Intact operation, first section outage as well as branch 6-7 and 53-54 outages were studied in summer minimum

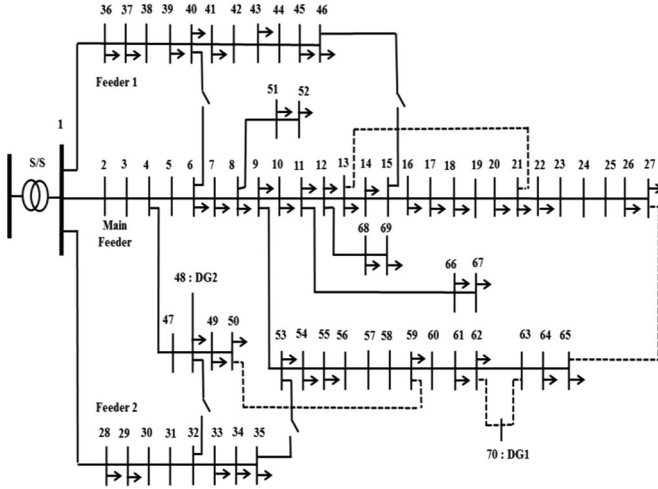


Fig. 8. Single line diagram of the 69-bus test system.

TABLE VI
TOTAL INVESTMENT COSTS FOR 69 BUS SYSTEM

Planning Period 1			Planning Period 2		
Sce.	Load Level DG1 Capacity	Inv. Costs (£)	Sce.	DG2 Capacity	Inv. Costs (£)
S1	LL: 100% DG: 100%	£168 812	S5	100%	£81 450
S2	LL: 100% DG: 150%	£71 169	S6	150%	£56 750
S3	DG: 100% LL: 80%	£71 222	S7	100%	£0
S4	LL: 80% DG: 150%	£180 374	S8	150%	£27 006
			S9	100%	£0
			S10	150%	£0
			S11	100%	£0
			S12	150%	£3 070

regime, whilst intact operation and first section outage were considered in winter peak regime.

A total of 12 scenarios (S1-S12) were investigated; 4 scenarios were considered in planning period 1 and then 2 scenarios in planning period 2 for each scenario from period 1, giving 8 scenarios in period 2 – Table VI. A load growth of 20% between year 5 (end of period 1) and year 10 (end of period 2) was assumed, as well as that a new 1 MVA DG2 unit is connected directly to node 48 in period 2. All scenarios S1-S12, scenario parameters and the total investment costs are tabulated in Table VI. Comparison of scenario S1 and S2 costs shows that the larger DG1 unit contributes significantly to the reduction of reinforcement costs in winter peak, because it supplies local loads and prevents excessive power transfers over other branches. In scenarios S2 and S3, roughly the same capital costs are required because both the load level and DG1 capacity have decreased in S3. However, DG size can become a great challenge in summer minimum regime, when the load is low. Scenario S4 depicts such a case in which load level is only 80% and DG1 injection is 150%, resulting in high DG-driven reinforcement costs in the summer minimum regime. In planning period 2, difference in S5 and S6 costs is again due to reduced branch power flows in winter peak regime when a ‘big’ DG2 unit is connected. In scenario

TABLE VII
ASSET CONSTRUCTION & REINFORCEMENT FOR 69-BUS SYSTEM

Scenario	Assets Construction and Reinforcements
S1	Branch: <u>27-65</u> , <u>62-70</u> , 1-2, 2-3, 3-4 Switch: <u>54-55</u> , 1-2 Decommissioning: 12-13
S2	Branch: <u>50-59</u> , <u>62-70</u> Switch: <u>6-7</u> , 1-2 Decommissioning: 53-54
S3	Branch: <u>27-65</u> , <u>62-70</u> Switch: <u>58-59</u> , 1-2 Decommissioning: 12-13
S4	Branch: <u>50-59</u> , <u>62-70</u> , 1-2, 2-3, 3-4 Switch: <u>4-47</u> , <u>6-7</u> , <u>44-45</u> , <u>9-53</u> , <u>48-49</u> , <u>50-59</u> , 1-2
S5	Branch: <u>50-59</u> , 48-49, 49-50 Switch: <u>62-63</u>
S6	Branch: <u>50-59</u> , 48-49 Switch: <u>47-48</u> , <u>63-64</u>
S8	Branch: <u>27-65</u> Switch: <u>64-65</u>
S12	Switch: 1-2

S7, winter peak investments were appropriate for the assumed load growth and DG2 was not big enough to cause ‘problems’; this resulted in zero capital costs required. However, when the DG2 size is increased (DG1 = DG2 = 150%), summer minimum regime becomes critical requiring construction of new assets in S8. Winter peak driven reinforcements in S3 were sufficient for the assumed load growth in period 2 and DG1 and DG2 sizes were not ‘critical’, so no asset interventions were required in scenarios S9 and S10. Finally for the 80% load level and DG1 = 150%, big investments in S4 were almost completely sufficient in planning period 2; zero costs were obtained in S11, whilst a new switch was required in S12. The zero cost scenarios S7, S9, S10 and S11 in planning period 2 can also be interpreted that DG2 power injection at node 48 was sufficient to balance-out the load growth in the studied 5-year period.

Details of capital investments for each scenario S1-S12 are listed in Table VII. Branch decommissioning and a circuit construction in a new corridor were part of the optimal solution in scenarios S1, S2 and S3. However, in S4 there was no suitable branch to be decommissioned that could satisfy operation constraints in all studied regimes, which resulted in construction of several new switchgear. Relatively very few asset reinforcements were required in period 2 scenarios other than S5, mainly because DG1 and DG2 outputs were sufficient to cover the load growth. Asset interventions in S8 and S12 are related to the summer minimum period. It should also be noted that replacement of switchgear 1-2 in S12 was due to the reverse power flow to grid. The highest voltage in planning period 2 was 1.043 at node 70 in summer minimum, intact network regime, whilst the lowest voltage was 0.95 at node 27 in winter peak, first section outage regime.

Reinforcements and decommissioning of branches is illustrated in Fig. 9 for all considered scenarios. Branch 62-70 was used to connect new DG1 to the network for all four scenarios in planning period 1. New branches 50-59 and 27-65 were part of

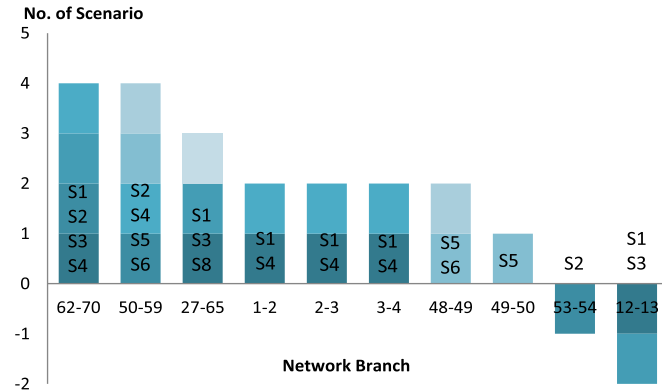


Fig. 9. Branch reinforcements and decommissioning for all scenarios.

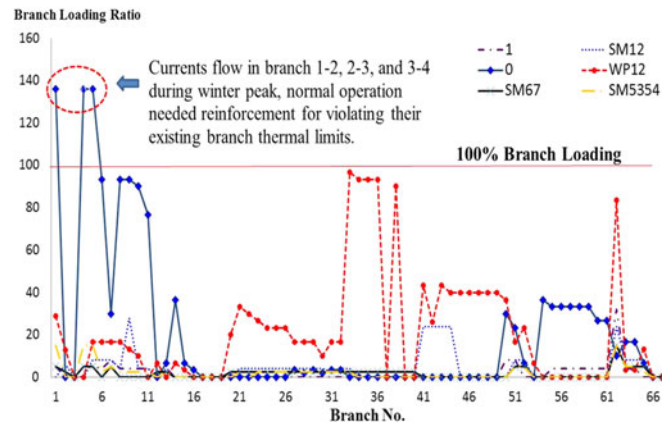


Fig. 10. Ratio of branch currents for all considered regimes for Scenario 1 of the 69-bus test system.

the optimal solution in several scenarios because they provide alternative routes of supply in emergencies.

Percentage ratios of branch current flows to their thermal ratings in the optimum solution are shown in Fig. 10 for scenario S1 for all considered regimes. The highest branch loadings were on branches 1-2, 2-3, and 3-4 during winter peak regimes. Loading of all branches is low for summer minimum regimes as the load is small and DG injection at node 70 helps to supply neighboring nodes. On the other hand, construction of new branch 50-59 as well as new switches in branches 4-47, 6-7, 44-45, 9-53, 48-49, and 50-59 were triggered by reverse power flows caused by DG1 unit in S4, as the load was low (80%) and DG1 injection was high (150%).

C. Case Study III: 119 Bus Test System

The 119-bus test system is employed to evaluate model performance in terms of computation time and memory size on a large distribution system. Single line diagram of the test system is shown in Fig. 11 with network parameters defined in [65]. The *intact* network is highly loaded, so to enable realistic modelling of single outages, winter peak loads were set to fifty percent of the demand shown in [65]. New DG unit at node 120 can be connected either through branch 36-120 or 37-120. Intact operation, first section outage as well as branch 7-8 and

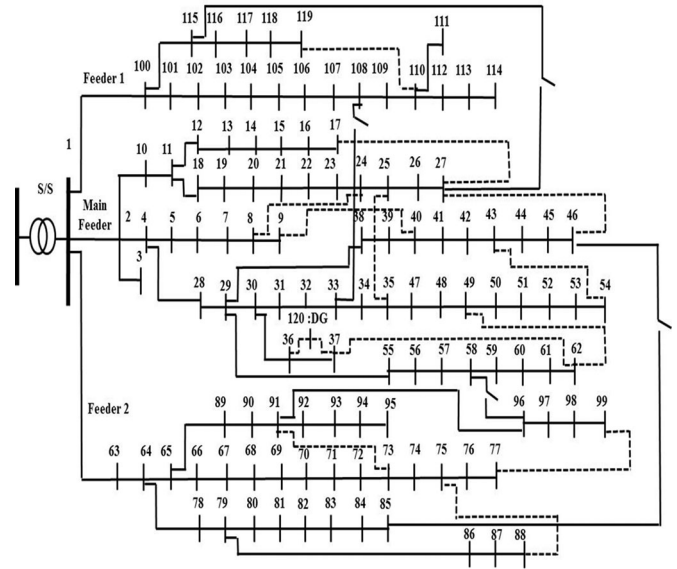


Fig. 11. Single line diagram of the 119-bus test system.

TABLE VIII
TOTAL INVESTMENT COSTS FOR 119-BUS SYSTEM

Investment Type	Investment Cost (£)
New Branch Construction	9-40, 37-120
Existing Branch Reinforcement	1-100
Branch Decommissioning	4-28
Total Investment Costs	£ 151 808

8-9 outages were considered for winter peak load, while intact and first section outage were considered for summer regime.

A summary of the investment costs needed to ensure secure operation of the network is presented in Table VIII. The DG is optimally connected to the network via new branch 37-120, whilst construction of new branch 9-40 enabled resupply of neighboring loads during section outages. Reinforcement of existing branch 1-100 was also required in case of the first section outage in winter regime, as the resupply was done via 'Feeder 1'. Decommissioning of branch 4-28 enables radial configuration with no constraint violation in all operating regimes with minimum cost.

Voltage magnitudes of network buses are plotted in Fig. 12 for all studied regimes. The highest voltage was 1.02 pu at DG node 120 in winter peak, first section outage, whilst the lowest voltage was 0.94 at node 46 in winter peak, outage of branch 8-9, as the supply was restored through single supply point 33-108; note that nodes 28, 29, etc. do not have very low voltages because the DG unit is very close.

VIII. COMPARATIVE STUDIES

The proposed models are compared against the traditional approach, whereby the intact network and contingencies are studied separately in turn, decommissioning is not allowed, and reinforcement results are accumulated. The developed optimization model (1)–(27) with a single regime k was used for this

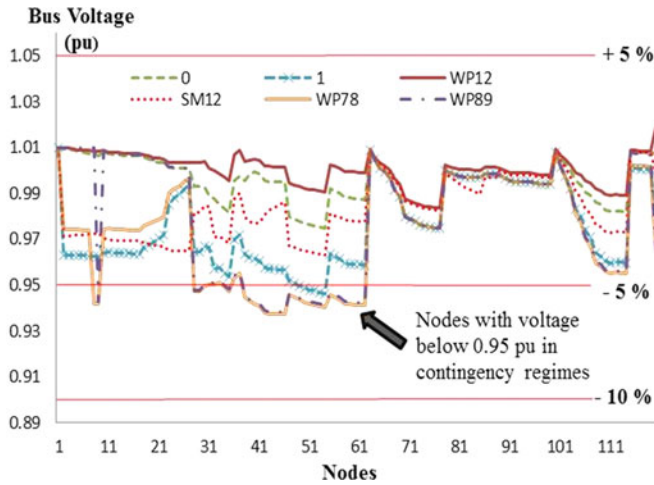


Fig. 12. Voltage profile of the 119-bus test system.

TABLE IX
COMPARISON OF COSTS: TRADITIONAL VS PROPOSED MODEL

Model/Test System		33 Bus System	69 Bus System	119 Bus System
Traditional Approach		146,782	269,198	247,400
Simplified Model		163,730	374,548	225,090
Full Model		141,677	215,452	151,808
Cost Difference	Traditional	5,105	53,746	95,592
	Simplified	22,053	159,096	73,292

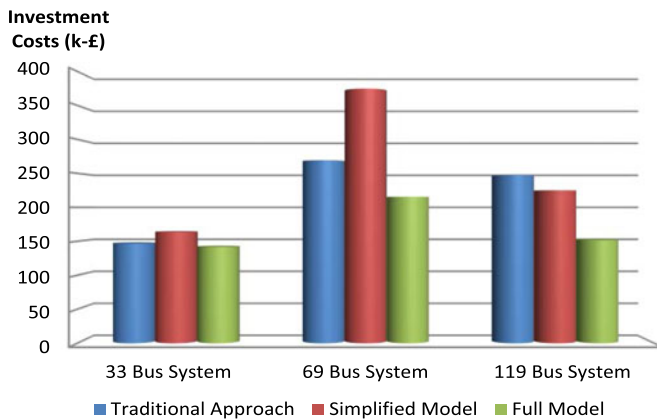


Fig. 13. Costs comparison between traditional approach and proposed models for all test networks.

purpose. A summary of non-discounted costs is presented in Table IX and Fig. 13; the first row shows results of the utility planning method, the second of the simplified model (1), (28)–(44), and the third of the full model (1)–(27), which correspond to scenario S1 from Table VI with 100% load level and 100% DG1 capacity. Lower investment costs were observed for all test systems when comparing the full model with the traditional approach. In case of the 33-bus system, cost reduction of $\sim 4\%$ was achieved due to the requirement to construct one switchgear less when studying all outages simultaneously. In case of the 69-bus system, the optimal solution of the full model did not

TABLE X
SENSITIVITY STUDY RESULTS

Cost Type	Base Case	Backfeeding:Feeder 1	DG Connect.:Firm
Total Costs (£)	215,452	392,082	219,912
Costs Difference (£)		176,104	4,460
% Difference		81.5%	2.07%

contain construction of branch 50-59 and three new switchgears, which otherwise would be needed if applying the traditional approach. A cost saving of $\sim 25\%$ was achieved in this way. Similarly in case of the 119 bus system, there was no need to construct branches 8-24 and 100-101, which resulted in cost savings of $\sim 63\%$.

On the other hand, a meaningful comparison between the full and the simplified model can be done by assuming that branches with an existing switch in the full model can only be used in the simplified model; note that decommissioning of circuits and installation of new switchgear is not allowed in the simplified model. This resulted in much higher investment costs, which were in the range $\sim 16\%$ up to $\sim 74\%$. The results show that relatively small investments in new switchgear can prevent high reinforcement costs of existing branches.

A. Sensitivity Analyses

Sensitivity studies were carried out on the 69-bus system with scenario S1 data (Table VI). Table X gives a comparison between the total non-discounted investment costs resulted from reducing the number of allowable back-feeding feeders to one and increasing the number of DG connecting branches to two. Reduction in the number of back-feeding feeders resulted in high currents through five branches between buses 1 and 40, as the supplies were restored through branch 6-40; these five branches and the existing switchgear in branch 1-36 had to be reinforced. This has led to sharp increase in the total investment costs of $\sim 81.5\%$ compared to the base case.

A firm connection of DG1 unit at node 70 via new branches 25-70 and 48-70 (not shown in Fig. 8) resulted in DG power injection to a larger section of the network. The additional cost of having the *firm* DG connection was compensated by not having to construct branch 27-65 from the ‘non-firm’ optimal solution; the overall cost change was $\sim 2\%$.

IX. CONCLUSION

This paper presents a flexible integrated methodology for distribution planning based on the UK utility concepts. Overall problem formulation, investment and operation models are presented; the focus is put on the investment optimization which gives optimal reinforcements of the MV network and DG connections. The proposed investment model considers explicit modelling of the network security in conjunction with multiple operating regimes, new switchgear installation and branch decommissioning, as well as company specific rules for customer connection and supplies restoration. The proposed modelling of

uncertainties with the aid of the probabilistic decision tree offers a flexible concept that can include any number of uncertainties, as well as easy interpretation of results.

The studies done on three real-sized MV networks have shown several advantages of the developed methodology. Impacts of load and DG uncertainties were studied on 33-bus and 69-bus test networks. The 33-bus test network is demand dominated and significantly higher investment costs are required for forecast higher load levels – ratio of $\sim 5/1$ was obtained in planning period 1. It is also shown on this example that 39% reduction in total investment cost is possible if the horizon/regulatory year planning method is applied instead of minimum cost method at all stages. The proposed investment model is then compared with the well-established model [17] and it has been shown that 12% reduction in total investment costs can be obtained due to explicit modelling of security constraints. Studies of load and DG uncertainties on the 69-bus test network have shown both positive and negative impact of DG on total investment costs. If the DG size is increased by 50% in planning period 1, a reduction of 57.8% in investment costs can be achieved. On the other hand when the load level is lower, the same increase in DG capacity in planning period 1 can lead to 153.2% increase in investment costs; this is the consequence of DG driven investments. Comparative studies between the proposed models and the traditional approach were also done on the 69-bus test network. It was found that up to 63% reduction in investment costs can be achieved, because of explicit modelling of critical outages and multiple operating regimes, as well as branch decommissioning. Besides, comparison of the full and the simplified proposed investment models has clearly shown benefits of constructing new switchgear and branch decommissioning, since reduction in investment costs can go up to 74%. The importance of applying ‘proper’ rules for supplies restoration is evidenced in sensitivity studies, from which it can be observed that limiting the backfeeding to one adjacent feeder has resulted in capital expenditure increase of 81.5%. Finally, robustness and applicability of the proposed model is demonstrated through studies on the 119-bus test system.

Current investigations are focused on the operational problem, whose outlines are presented in Section VI; the main findings will be reported in a separate paper.

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